NAME: _____ **CLASS:** _____

Extended Experimental Investigation

Investigating HEAT Conductance

TASK: To investigate (and therefore complete this worksheet):

- The relationship between the type of material and its ability to conduct heat.

Background What is the theory behind your investigation? What makes different materials conduct heat better or worse?	
Aim To determine the …	
Hypothesis (a testable prediction with explanation). What (and why) do you expect to happen?	
Materials (Bulleted list of equipment. Use a double column format)	
Risk assessment What are the possible hazards and how will you manage them?	

Source of risk	Potential risk	Degree of risk	Safety precautions

Method

What were the steps in the investigation?

List these in **numbered steps.**

Represent apparatus with a diagram

Results and observations

If you do three trials for each variation in your dependant variable, record temperature and times in your notebook and only put the averages in Table 1

Put a heading for your table

Table 1: _____

Do three trials each time you vary the independent variable

Record measurements for in a table.

Type of Material	Trial	Temperature (°C)							Average △Temp (°C)	Rate of Temp change (°C/min)
		0 min	2 min	4 min	6 min	8min	10min	△Temp		
	1									
	2									
	3									
	1									
	2									
	3									
	1									
	2									
	3									
	1									
	2									
	3									
	1									
	2									
	3									

Show any sample calculations you used to fill in the results table

Draw a graph of Temperature versus time for each material.

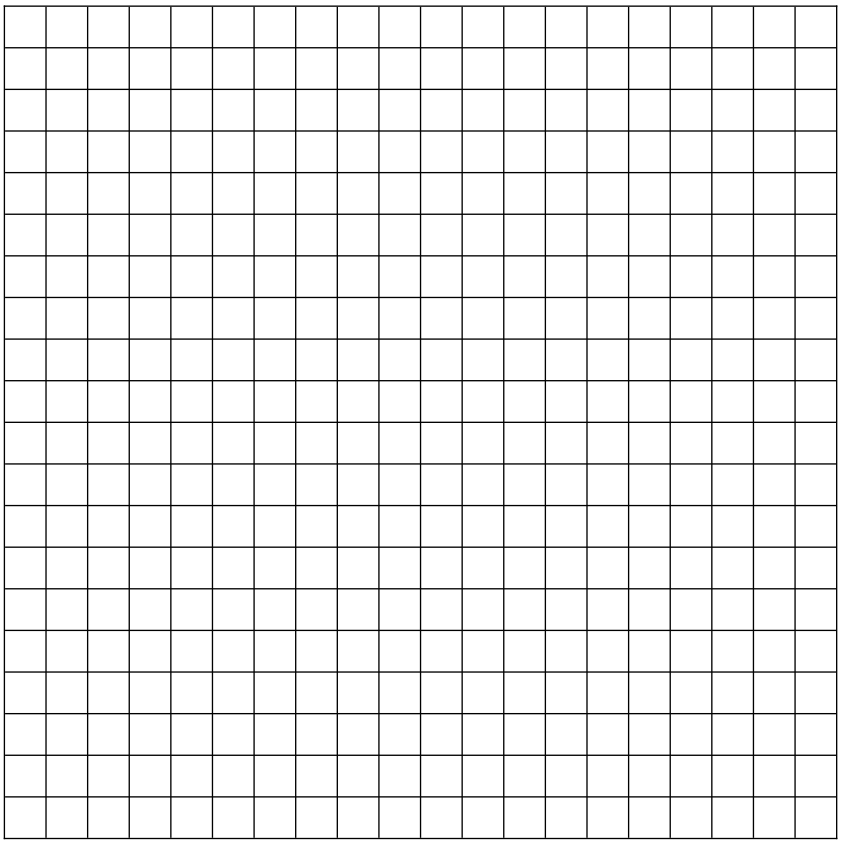
Choose the “best” (most representative) trial to graph for each material.

There will be four or five lines (one for each of the materials you investigated.

You will need a key explaining which line is for which material.

Draw a **trend line** for each set of dots (don’t simple join the dots together0

Graph 1: Temperature versus time for each material

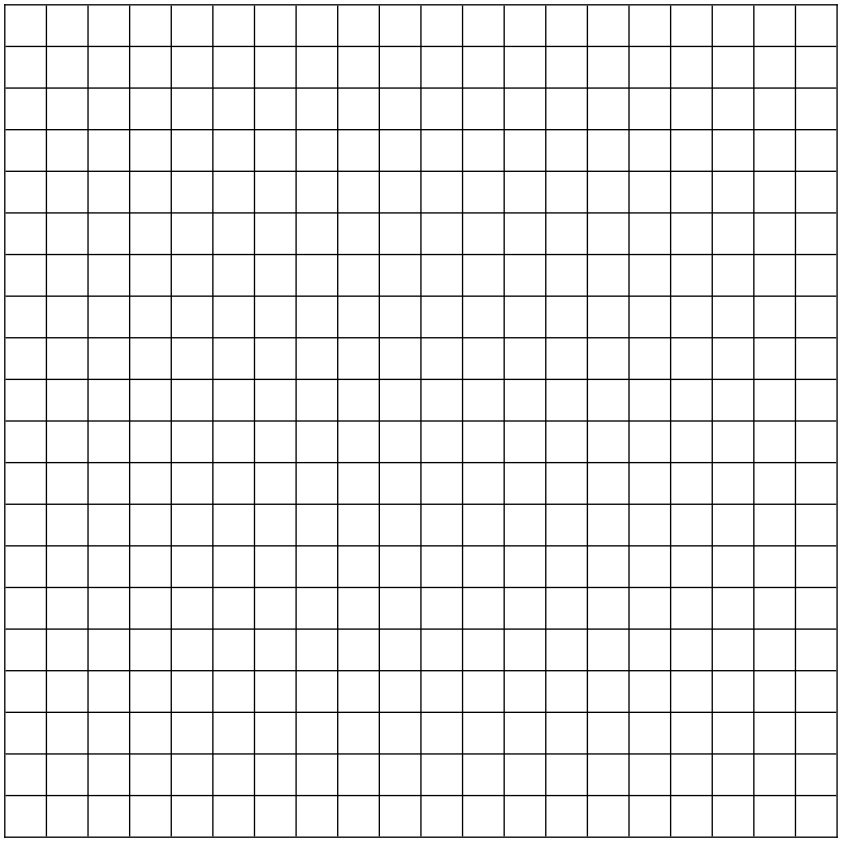


Graph 2: Rate of temperature change for each material

Draw a bar graph of rate of temperature change for each material.

This uses column 11 and column 1 from your table.

A bar graph is simple a picture of your results so it is not very useful to scientists, and therefore not used very often. However a bar graph does show your results very clearly



Analysis of

results

What can you decide or **conclude** from you results? How does the data support this?

Write a paragraph on each conclusion you make. Use a conclusion as your topic sentence for each paragraph, elaborate or explain it if necessary, then provide the evidence in the form of data from your results.

Identify any errors which are apparent in your results **AND** explain why you think this.

There are two ways to judge errors and you should use both to identify any possible error.

Firstly- Do the points on your first graph make a consistent trend (are the points close to the trend line or a little "scattered"?).The closer the points to the trend line the less likely error occurred.

Secondly – Is your answer correct? If you have some final value or equation (model), you can make a judgement as to whether it is correct. Compare it with the expected result or your common sense? If it is not correct you have error?

Conclusion

State your conclusion(s) and the implication of them (elaborate on each conclusion with a couple of extra sentences). For example, explain why some materials cause a lower rate of heat loss than other materials.

There may be 1 main conclusion or several.

State the degree of error in your conclusion(s) and provide reasons the error has occurred. Provide suggestions to improve the experimental design so these errors could be avoided. Is there so much error that your conclusions are not valid?

Sum up your conclusion by stating if your main conclusion agreed with your hypothesis, and if your conclusions are valid or not.